NYC Yellow Taxi Analysis

Lynx Use Case – Data Analyst

LYNX

Agenda

- Executive Summary
- Trips General Profile
- Trip Segmentation
- Demand Evolution
- Tip Analysis
- Data Quality
- Q&A

Executive Summary

This assignment analyzes TLC NYC Yellow Taxi trip records from 2018 through January 2023, providing a multi-year view of trip demand, revenue patterns, and operational behavior across the city. The analysis focuses on identifying key commercial trends, segment differences, and data quality considerations that are relevant for business reporting and decision-making.

From a business perspective, the main takeaway is that the taxi market is not homogeneous:

- It is composed of distinct commercial segments with different behaviors, economics, and operational needs.
- Performance should therefore be interpreted through trip patterns, since location and timing have a direct impact on both volume and revenue.
- This is reflected in four clear segments: airport trips, Manhattan-Manhattan, Non-Manhattan-Manhattan, and Non-Manhattan-Non-Manhattan.

From a Data Quality perspective, there are also important issues to keep in mind:

- congestion_surcharge field appears to require further validation, and there may be additional concerns related to sensor reliability and the store-and-forward process.
- These elements should be reviewed carefully, as they may affect the integrity of KPI reporting and the robustness of any conclusions drawn from the data.

Trips General Profile

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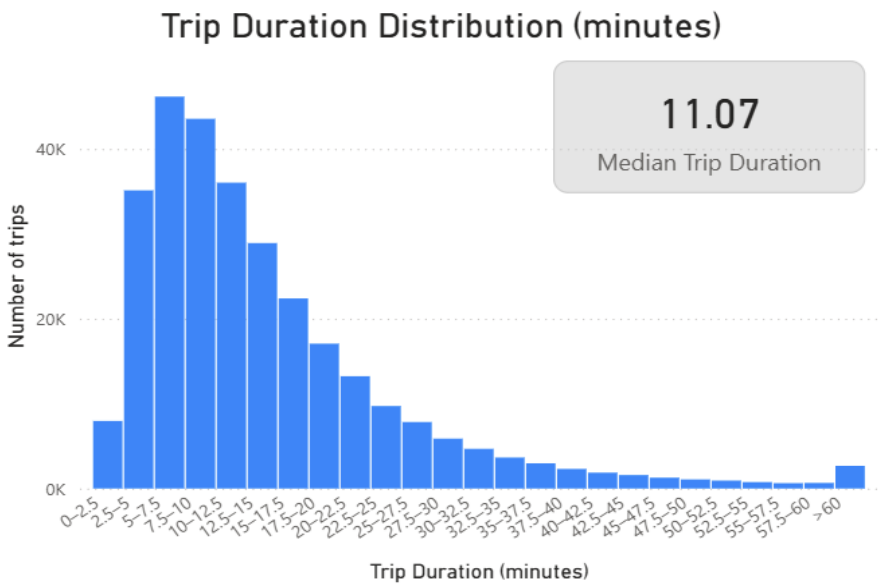
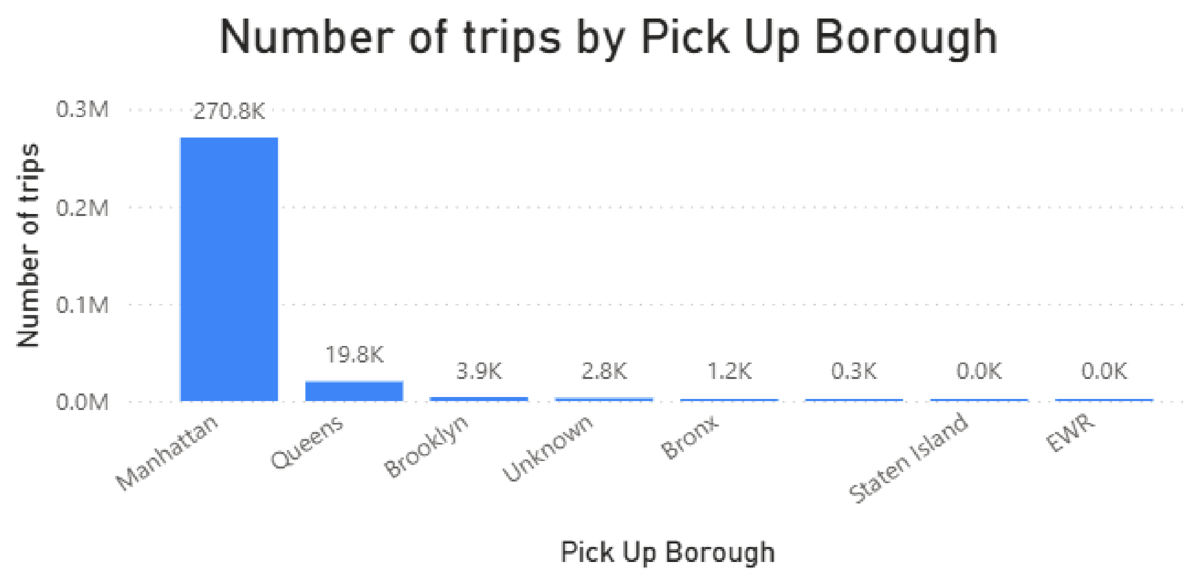
NYC taxi demand is concentrated in short Manhattan rides with clear weekday commuting peaks





Recommendation:

The core business is highly urban and time-sensitive, so driver allocation should prioritize Manhattan and peak weekday hours.

Trip Segmentation

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Airport trips are a distinct high-value segment with much longer duration, distance, and fare levels.

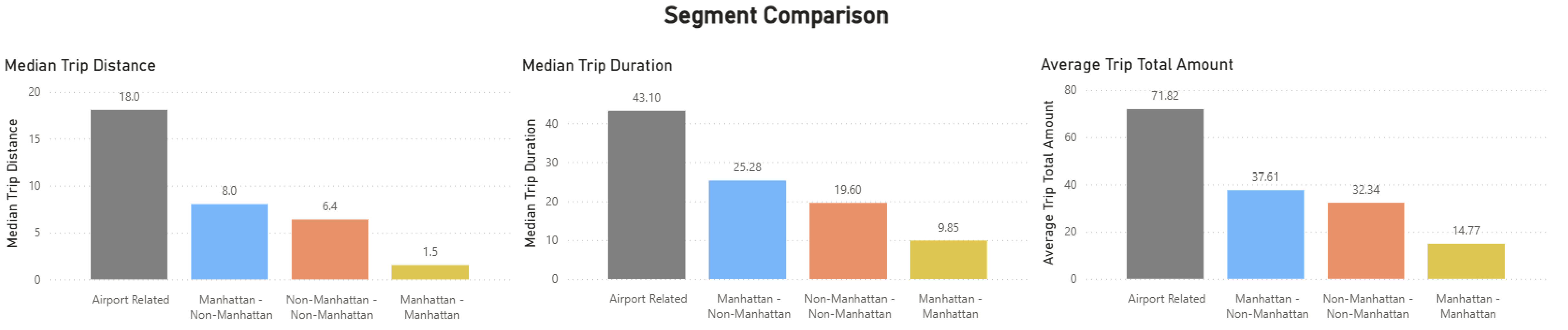
Segment	trips	median_distance	median_duration	avg_total_amount
Non-airport trips	291241	1.70	10.833333	17.611668
Airport-related trips	7583	18.04	43.100000	71.821208

Recommendation:

Airport trips should be reported separately and managed as a premium operational segment rather than mixed with general urban trips.

Trip Segmentation

Airport Related Trips, Manhattan-to-Manhattan, Manhattan-to-outer borough, and non-Manhattan trips behave differently.

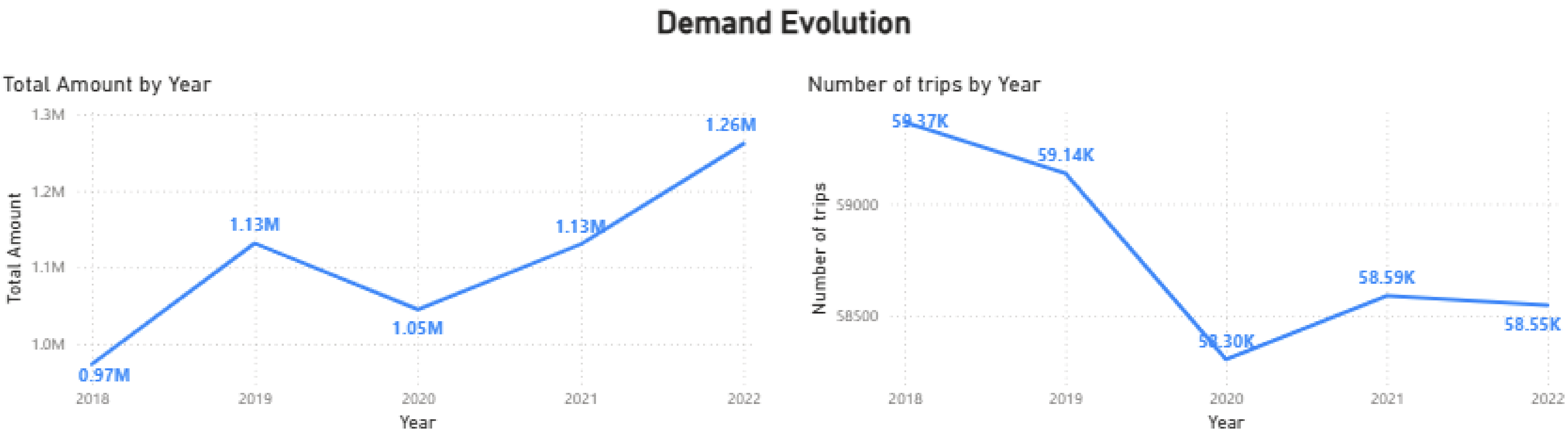


Recommendation:

Trip segmentation should be embedded in reporting to avoid masking distinct demand and pricing patterns.

Demand Evolution

Trip count has declined over time, while total amount has increased.

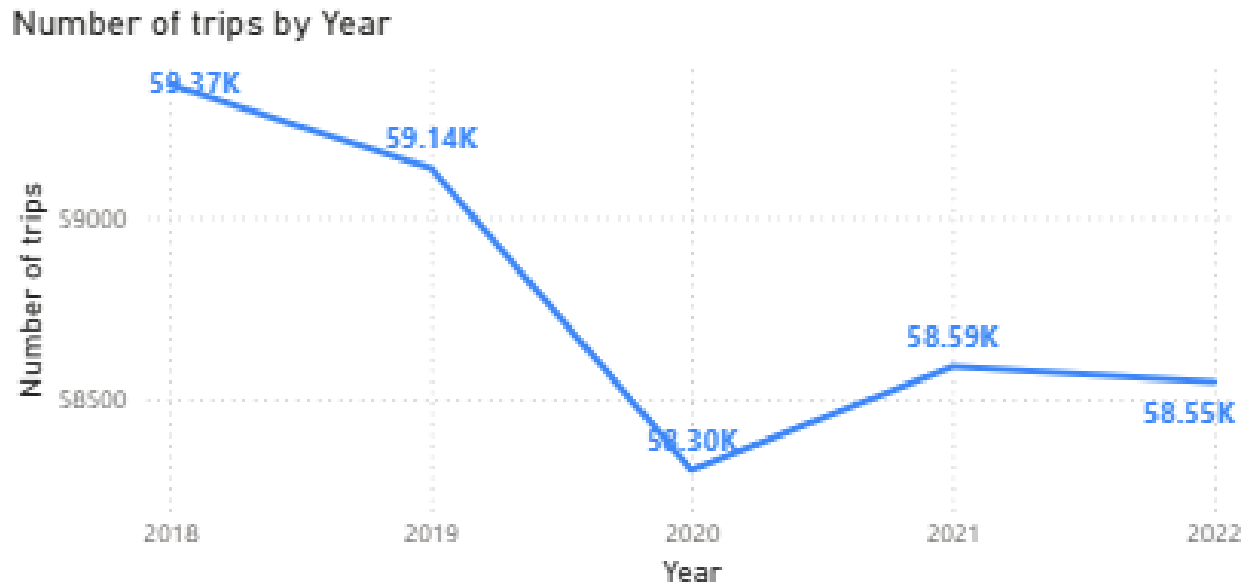


Recommendation:

Revenue growth is being driven more by trip distance and pricing than by stronger demand, so trip volume is the better KPI for market

Demand Evolution

The pandemic created a clear break in demand in 2020.



Recommendation:

Historical trend analysis should treat 2020 as an exceptional period, not a normal reference year.

Market Structure

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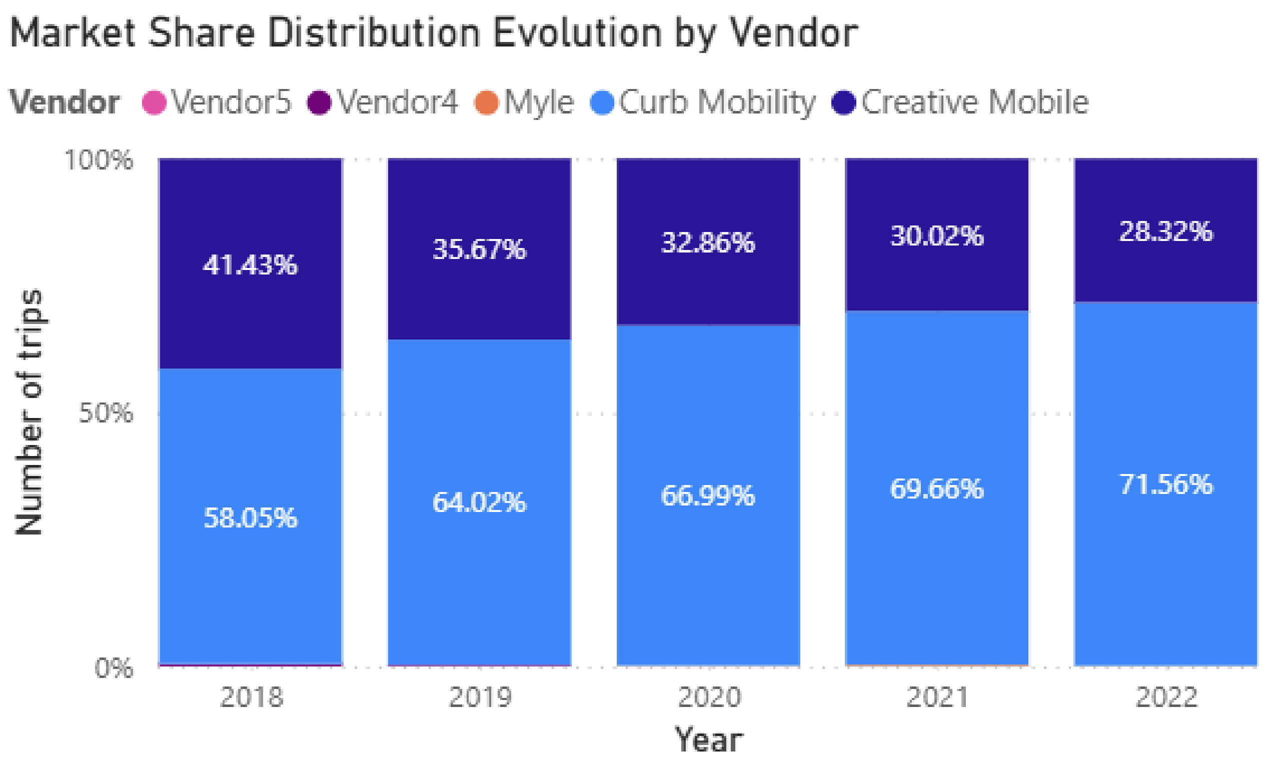
Vendor concentration is very high, with one dominant player consistency increasing its leadership and one main competitor.



Recommendation:

The market shows concentration risk, and competitive analysis should monitor whether dominance continues to increase.

Tip analysis

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Credit card trips show strong recorded tipping behavior, while cash tips are not captured.

	payment_name	tip_amount_average	tip_rate_average
0	Credit Card	3.02	25.24
1	Flex Fare	2.32	12.55
2	No Charge	0.00	0.00
3	Dispute	0.00	0.00

Recommendation:

Tip KPIs are incomplete by design, so cash trips should be excluded from tip analysis and results should be interpreted as partial rather than total market behaviour.

Data Quality

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The field `congestion_surcharge` appears inconsistently recorded, and the store-and-forward pattern suggests possible system issues.

Recommendation:
Data quality rules should be strengthened, and the surcharge and capture process should be reviewed before relying on these fields for revenue analysis.

Data Quality

Some extreme records may reflect sensor or capture problems rather than real trips.

Recommendation:
Outlier handling should remain conservative, and suspicious records should be monitored separately instead of being treated as normal business activity.

Q&A

Thank You!